

Berkay Benli

780-680-3211 | benli@ualberta.ca | berkaybenli.com | linkedin | github

EDUCATION

BSc Honors in Computing Science

University of Alberta, GPA: 3.4/4.0

Edmonton, AB

Sep. 2023 – May 2028

- **Relevant Coursework:** Algorithms I and II, Data Structures, Computer Architecture, Operating Systems, Advanced Mathematics, Databases (SQL), Software Engineering (Java)

EXPERIENCE

Software Developer Intern

Agriculture and Agri-Food Canada - Government of Canada

Edmonton, AB

Jan. 2026 – May 2026

- Developed **SLURM-based HPC pipelines** for **miRNA discovery** in *Pisum sativum* using **miRDP2**, **ShortStack**, and **miRador** across 48 **small RNA-seq samples**.
- Optimized **large-scale genomic workflows** by resolving **GPFS storage I/O** and **memory allocation bottlenecks** for improved pipeline performance and reproducibility.
- Created **Bash scripts** for **FASTQ preprocessing**, sample demultiplexing, and automated runtime monitoring across **multi-day HPC jobs**.
- Refactored **Nextflow DSL2 bioinformatics workflows** from global to per-entity state management using **branch operators** to support heterogeneous **genome assembly pipelines**.
- Debugged and updated pipelines for compatibility with **Nextflow 26** config parser v2, resolving **cross-scope variable** and configuration issues.

Computing Science and Mathematics Expert

Mercor AI

Edmonton, AB

Jan. 2025 – Sep. 2025

- Built **Python scripts** to automate rubric generation and streamline large-scale evaluation workflows, improving speed and consistency.
- Created lightweight **JavaScript prototypes** to simulate LLM frontend behavior, enabling rapid testing of rubric coverage and error detection.
- Evaluated **AI-generated STEM solutions** for factual accuracy, logical consistency, and clarity, providing structured feedback to improve model performance.

AI Prompt Engineer and Front-End Evaluator

Outlier AI

Edmonton, AB

May 2025 – Sep. 2025

- Engineered **natural-language prompts** to elicit functional front-end code responses from LLMs, focusing on **HTML/CSS/JS**.
- Led **multi-turn conversations** with models to iterate on UI/UX improvements, bug fixes, and new feature implementations.
- Authored **rubrics** to evaluate LLM output against clearly defined frontend requirements, emphasizing clarity, technical precision, and visual fidelity.

Software Team Member

UAS Starr Alberta Rocket Research

Edmonton, AB

Jan. 2024 – May 2024

- Enhanced rocket tracking accuracy by **12%** through the integration of efficient tracking algorithms, including **Kalman filtering**, and optimized code performance.

PROJECTS

Sample Searcher

React, Node.js, MySQL, Python, AWS

Link: [Samplesearcher.ca](https://samplesearcher.ca)

- Developed a **full-stack web application** that lets users browse and discover music samples from a curated dataset of 2,000+ songs scraped with Python.
- Created **RESTful API endpoints** with Node.js and Express, backed by a MySQL database for efficient data management.
- Deployed the backend on **AWS EC2** using **Docker**, **AWS CDK**, and CI/CD pipelines to ensure scalable, automated deployment.

TECHNOLOGIES

Languages: Python, C, Java, Swift, JavaScript, TypeScript, SQL, NoSQL, Assembly

Frameworks: React, Next.js, Node.js, Tailwind CSS, Vite.js, MongoDB, AWS CDK

Developer Tools: Git, Docker, Linux, AWS, Google Cloud, Heroku, Netlify